

Interfacial Charge Transfer and Morphology Govern Magnetism in Ultrathin VO_x/MoSe₂ Heterostructures

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Abstract

Engineering magnetism in two-dimensional semiconductors through interfacial interactions offers an attractive alternative to substitutional doping and provides opportunities for integrating magnetic and optoelectronic functionality within the same heterostructure. Here, we investigate ultrathin VO_x films deposited on monolayer MoSe₂ and demonstrate a pronounced enhancement of the magnetic response upon formation of the VO_x/MoSe₂ interface. Combined structural, spectroscopic, magnetic, and first-principles analyses support a consistent picture in which finite VO_x clusters possess intrinsic magnetic moments that are further enhanced by electron transfer from MoSe₂ to the oxide. The predicted charge transfer is also consistent with the increased trion contribution observed in the photoluminescence response of the heterostructure. Beyond this interfacial enhancement, the magnetic behavior is strongly governed by the morphology of the ultrathin oxide. A crossover from discontinuous clusters to more continuous films produces qualitatively different temperature dependences: the continuous-film regime exhibits conventional thermal demagnetization and coercive softening, whereas the clustered regime displays an unusual increase in magnetization and nonmonotonic coercivity with increasing temperature. These contrasting responses are captured by a phenomenological model in which morphology determines the collective magnetic dynamics through a distribution of effective reversal barriers and their thermally activated accessibility. Together, the results establish a unified picture in which interfacial charge transfer enhances the local magnetic moments of VO_x, while nanoscale morphology governs their collective magnetic dynamics, providing complementary routes for controlling magnetism in oxide/transition-metal dichalcogenide heterostructures.

Keywords: oxide/TMD heterostructures, two-dimensional materials, interfacial magnetism, interfacial charge transfer, thickness-dependent magnetism.

The ability to engineer magnetic functionality in atomically thin materials has emerged as a central objective in condensed matter physics and nanotechnology because of its potential impact on spintronic, magneto-optical, and quantum information devices.¹⁻³ Two-dimensional transition-metal dichalcogenides (TMDs) are particularly attractive owing to their strong spin-orbit coupling, direct band gaps in the monolayer limit, robust excitonic effects, and compatibility with van der Waals heterostructures.⁴⁻⁷ Considerable effort has therefore focused on introducing magnetic functionality into otherwise nonmagnetic TMDs while preserving their structural and optical quality.^{8, 9}

Several strategies have been explored to realize magnetism in TMDs, including substitutional magnetic doping,¹⁰⁻¹⁷ defect engineering,^{18, 19} proximity coupling to magnetic materials,²⁰⁻²² and intrinsically magnetic van der Waals crystals.^{2, 3, 23-25} Although these approaches have produced important advances, they can introduce limitations such as chemical disorder, degraded carrier transport, limited ordering temperatures, or restricted material choices. These challenges motivate alternative routes for modifying magnetic properties without substantially disrupting the host semiconductor lattice.²⁶

Interface engineering has recently emerged as one such alternative.²⁷⁻²⁹ Electronic charge redistribution across oxide/semiconductor interfaces can modify orbital occupation, oxidation state, and exchange interactions without requiring extensive atomic interdiffusion.³⁰ Transition-metal oxides are particularly attractive because their multiple oxidation states and correlated electronic structure make their magnetic properties highly sensitive to their local electronic environment.³¹⁻³³ Among transition-metal oxides, vanadium oxides are especially versatile because vanadium can adopt multiple oxidation states, while the V–O system supports numerous stoichiometric, oxygen-deficient, mixed-valence, and metastable phases with diverse crystal structures and electronic properties.³⁴ While bulk V_2O_5 is generally nonmagnetic, reduced coordination, oxygen vacancies, finite clusters, and non-stoichiometric VO_x nanostructures can develop localized magnetic moments.³⁵⁻³⁷ Recent first-principles calculations on a V_2O_3 /WSe₂ van der Waals heterostructure further illustrate this potential, predicting strong interfacial charge redistribution and magnetic proximity coupling between a two-dimensional vanadium oxide and a TMD semiconductor.³⁸ Consequently, ultrathin VO_x deposited on a two-dimensional semiconductor provides an excellent platform for investigating how interfacial electronic

interactions modify pre-existing magnetic moments while preserving the optical and electronic properties of the underlying 2D semiconductor.

Motivated by our preliminary experimental results,³⁹ Woods' group employed density-functional-theory calculations to investigate an idealized single-crystal $\text{V}_2\text{O}_5/\text{MoSe}_2$ interface, predicting that interfacial charge transfer, orbital hybridization, and spin redistribution can promote ferromagnetism.⁴⁰ Together with recent theoretical predictions for $\text{V}_2\text{O}_3/\text{WSe}_2$ heterostructures,³⁸ these studies highlight the potential of vanadium-oxide/TMD interfaces for engineering magnetic functionality. However, the relationship among charge transfer, optical response, morphology, and magnetic behavior in experimentally realized nonstoichiometric VO_x/TMD heterostructures remains largely unexplored. Moreover, the role of oxide morphology—particularly the transition from isolated nanoclusters to continuous ultrathin films—in determining collective magnetic behavior remains poorly understood.⁴¹

In this work, we investigate ultrathin $\text{VO}_x/\text{MoSe}_2$ heterostructures in which the oxide morphology evolves from discontinuous islands to laterally connected ultrathin films with increasing thickness. By combining atomic force microscopy, Raman and photoluminescence spectroscopy, X-ray photoelectron spectroscopy, vibrating-sample magnetometry, first-principles calculations, and a phenomenological description of the temperature-dependent magnetic response, we establish a coherent physical picture. Our results indicate that finite VO_x clusters possess intrinsic localized magnetic moments that are enhanced through interfacial electron transfer from the MoSe_2 substrate. Independently, the oxide morphology governs the collective magnetic dynamics by modifying the distribution of effective reversal barriers and magnetic interactions, producing qualitatively different temperature dependences for clustered and continuous films. This unified framework links the structural, optical, electronic, and magnetic properties of the heterostructure and illustrates how interface engineering can be exploited to tailor magnetic functionality in oxide/semiconductor van der Waals systems.

Results and Discussion

Monolayer MoSe_2 samples were synthesized on Si substrates, covered with a 285 nm SiO_2 layer, using the water-assisted chemical vapor deposition (CVD) approach developed by our group.⁴²⁻⁴⁴ The process is schematically illustrated in Figure 1a; additional synthesis details are

provided in the Experimental Section. After CVD growth of MoSe₂, the samples were transferred to a high-vacuum thermal evaporator, where ultrathin VO_x films were deposited on the MoSe₂ surface (Figure 1b). The nominal VO_x thickness was monitored using an in-situ quartz crystal microbalance (QCM).

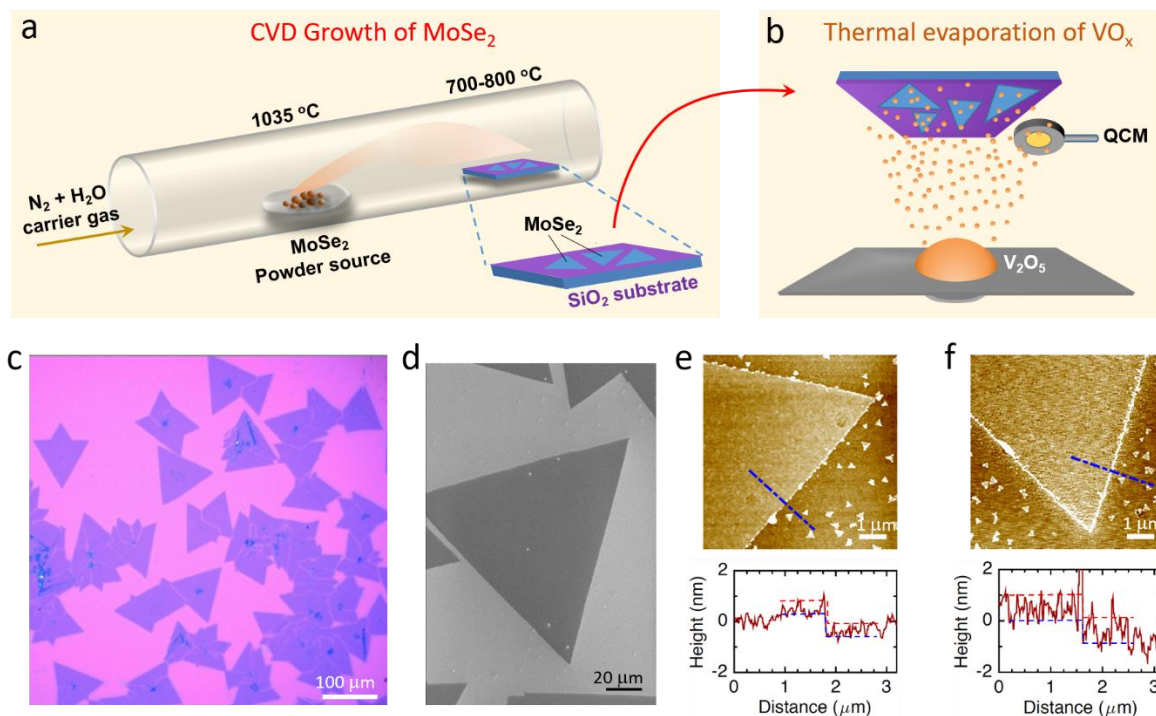


Figure 1. (a) Chemical vapor deposition of monolayer MoSe₂. (b) Thermal deposition of ultrathin VO_x films. (c, d) Optical and SEM images, respectively, of the CVD-grown MoSe₂. (e, f) Atomic force microscopy (AFM) images (top) and height profiles (bottom) of MoSe₂ before and after deposition of a nominal 0.3 nm VO_x layer, respectively.

Optical microscopy (Figure 1c) and scanning electron microscopy (SEM) (Figure 1d) reveal triangular CVD-grown MoSe₂ islands with a relatively high surface density. An atomic force microscopy (AFM) image and height profile of pristine MoSe₂ are shown in Figure 1e. The measured thickness is approximately 0.7 nm, consistent with monolayer MoSe₂, and the average roughness is below 0.45 nm, comparable to that of the SiO₂ substrate. After deposition of a nominal ~0.3 nm VO_x layer, as determined by the QCM, the average roughness approximately doubles (Figure 1f). The resulting room-temperature-deposited oxide is therefore consistent with a discontinuous nanoparticle-like morphology.

Using a shadow mask, step-like VO_x films with different nominal thicknesses were deposited directly on SiO_2 substrates (Figure 2). *Ex situ* AFM step-height measurements performed in tapping mode under ambient conditions yielded values slightly larger than the nominal QCM thicknesses (Figure 2e). The near-unity slope and small 0.3 nm intercept suggest a systematic offset that may arise from material-dependent apparent-height effects, because tapping-mode AFM step heights across dissimilar materials can depend on tip-sample interactions, capillary forces, adsorbed water layers, and measurement parameters.⁴⁵⁻⁴⁸ Accordingly, QCM values are used throughout this work as the nominal film thickness. The AFM analysis further indicates a thickness-dependent morphological evolution of the vanadium oxide layer. At the lowest nominal thickness, the greater roughness of the VO_x/SiO_2 region, relative to bare SiO_2 , is consistent with an island-like or discontinuous oxide morphology. With increasing nominal thickness, the roughness decreases towards that of the substrate, suggesting a transition to a more continuous film. This morphological evolution is relevant to the magnetic response discussed below: discontinuous VO_x clusters in the ultrathin regime exhibit finite-size and thermally activated behavior, whereas thicker, more continuous films are expected to approach the response of an extended oxide layer.

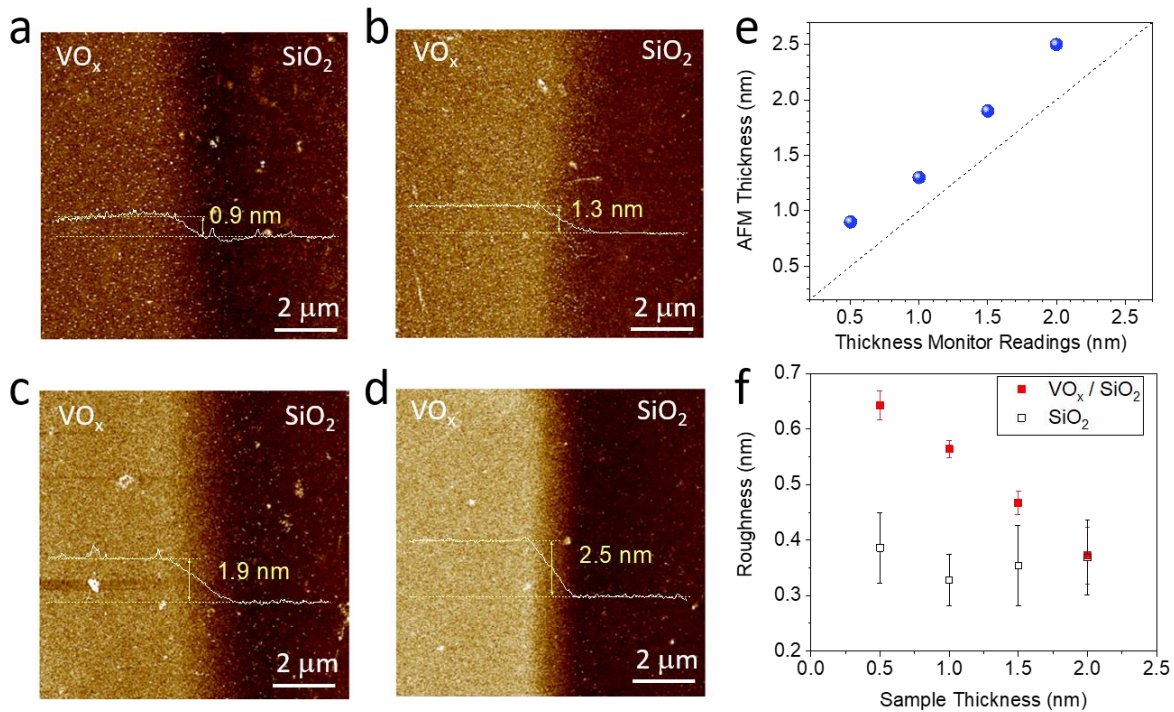


Figure 2. (a - d) AFM topography images and corresponding averaged step profiles for VO_x films deposited on SiO₂ substrates. (e) Correlation between film thickness measured by AFM and the nominal thickness measured by the in-situ QCM. (f) Surface roughness as a function of nominal film thickness determined from the AFM images in panels a-d.

Next, we compare the optical properties of pristine MoSe₂ with those of a heterostructure containing a nominal 0.7 nm VO_x overlayer. The Raman spectra are shown in Figure 3a. Two prominent peaks near 239 and 249 cm⁻¹ are assigned to the A_{1g} and 2E_{2g}²(M), respectively.⁴⁹ Additional weaker features at 146, 170, 286, 305, 319 and 361 cm⁻¹ are assigned to E_{2g}¹-LA(M), E_{1g}, E_{2g}¹, E_{1g}+LA(M), 2B_{2g}(M) and A_{2u}², respectively, following Nam et al.⁴⁹ The close similarity in peak positions and relative intensities before and after VO_x deposition indicates that the overlayer produces no readily detectable change in the crystalline structure of the underlying MoSe₂.

The photoluminescence (PL) spectrum of pristine MoSe₂ exhibits an intense peak at 1.53 eV (Figure 3b,c), consistent with our previous reports.⁴²⁻⁴⁴ The spectrum is slightly asymmetric toward lower energy and can be fitted with two Gaussian components (Figure 3c): a dominant neutral-exciton peak at 1.53 eV and a weaker trion contribution near 1.48 eV. After VO_x deposition, the PL maximum red-shifts slightly and the low-energy side broadens. For the nominal 0.7 nm VO_x/MoSe₂ sample, the red shift is less than 15 meV. Such a shift could arise from tensile strain^{50, 51} or a modified dielectric environment.⁵² Because the Raman spectra show negligible strain-related changes, the PL red shift is more plausibly associated with the altered dielectric environment produced by the VO_x overlayer.

The nominal 0.7 nm VO_x/MoSe₂ sample likewise exhibits an asymmetric PL peak that can be fitted with two Gaussian components (Figure 3d), centered at 1.515 and 1.475 eV and assigned to neutral excitons and trions, respectively. In both pristine and coated MoSe₂, the neutral-exciton–trion energy separation is 40-50 meV, close to the room-temperature value reported previously.^{53, 54} In addition, the trion-to-neutral-exciton intensity ratio (I_t/I_x) increases from 0.14 in pristine MoSe₂ to 0.37 after VO_x deposition. This increase could arise from charge transfer between the VO_x overlayer and MoSe₂ or from V incorporation into the MoSe₂ lattice; substitutional V at Mo sites is expected to induce p-type doping.¹⁰ Either mechanism would alter the free-carrier concentration and, consequently, the trion population. The XPS results below allow these possibilities to be distinguished.

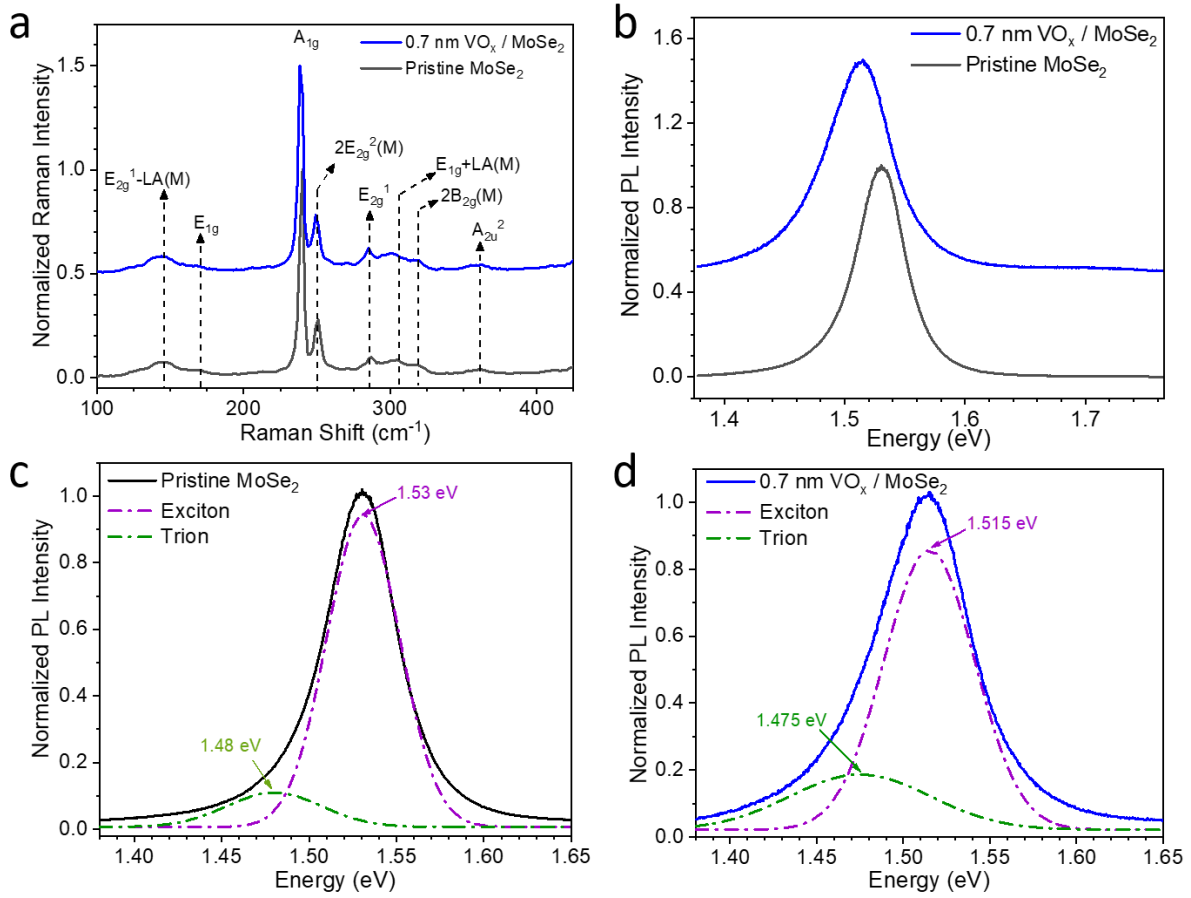


Figure 3. (a, b) Raman and PL spectra, respectively, of pristine MoSe₂ and nominal 0.7 nm VO_x/MoSe₂ heterostructure. (c, d) PL spectra of pristine MoSe₂ and 0.7 nm- VO_x/MoSe₂, respectively, showing the fitted neutral-exciton and trion contributions.

X-ray photoelectron spectroscopy (XPS) was used to assess whether V deposition produces measurable interdiffusion or substitutional incorporation into the MoSe₂ layer. As shown in Figure 4a, the Mo 3d and Se 3d spectra of pristine MoSe₂ and VO_x/MoSe₂ are nearly identical, with no significant core-level shifts or obvious changes in line shape. Previous work reported that even ~1% substitutional V doping in MoSe₂ can induce binding-energy shifts of approximately 0.5 eV in the Mo 3d and Se 3d levels.¹⁰ The absence of comparable shifts in our samples indicates that substitutional V incorporation is negligible or below the detection limit of the XPS measurements. In contrast, the V 2p spectrum of the oxide layer contains contributions from V⁵⁺ and reduced V species, indicating a deviation from ideal V₂O₅ stoichiometry. These results support a

heterostructure consisting of MoSe₂ interfaced with a mixed-valence, nonstoichiometric VO_x overlayer, rather than a significantly V-doped MoSe₂ layer.

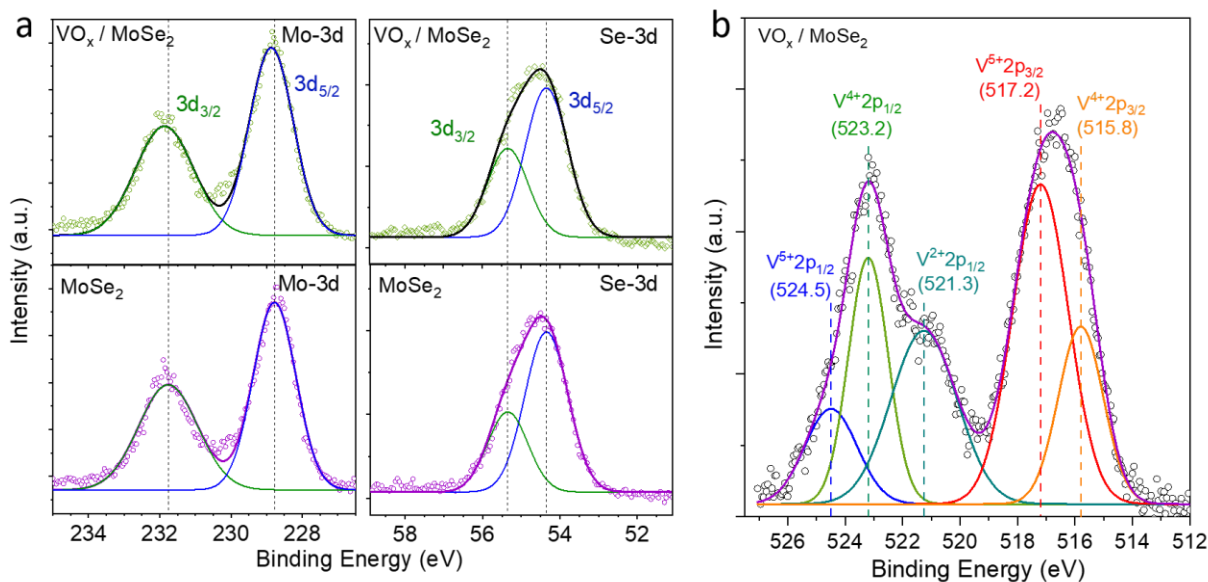


Figure 4. XPS spectra of (a) the Mo 3d and Se 3d regions for pristine MoSe₂ and VO_x/MoSe₂ and (b) the V 2p region for VO_x/MoSe₂.

The magnetic properties of the VO_x/MoSe₂ heterostructures were investigated using the vibrating sample magnetometer (VSM) option of a Quantum Design Physical Property Measurement System (PPMS). Figure 5 summarizes the areal magnetization as a function of magnetic field (M-H) after subtraction of the diamagnetic background. The top and bottom panels correspond to magnetic fields applied in-plane and out-of-plane, respectively. Pristine MoSe₂/SiO₂ and VO_x/SiO₂ show no detectable hysteretic response for an in-plane field (Figure 5a), whereas a weak out-of-plane response appears for the VO_x/SiO₂ films thicker than 1 nm (Figure 5d). In contrast, all VO_x/MoSe₂ heterostructures exhibit clear hysteresis, with a substantially stronger response in the out-of-plane configuration.

The enhanced ferromagnetic response observed after depositing vanadium oxide on MoSe₂ suggests that the interface actively modifies the magnetic properties of the oxide layer. To gain microscopic insight into this enhancement, we performed first-principles DFT+U calculations on representative VO_x nanoclusters (V₄O₁₀ and V₁₀O₂₂) interacting with a MoSe₂ monolayer; computational details and additional results are provided in the Supporting Information. The

calculations represent the early stages of film growth, in which isolated oxide clusters dominate before coalescence into a more continuous film, consistent with the AFM observations. Although only representative cluster geometries were considered, they capture local electronic interactions at the $\text{VO}_x/\text{MoSe}_2$ interface and provide a microscopic framework for interpreting the experimentally observed magnetic enhancement.

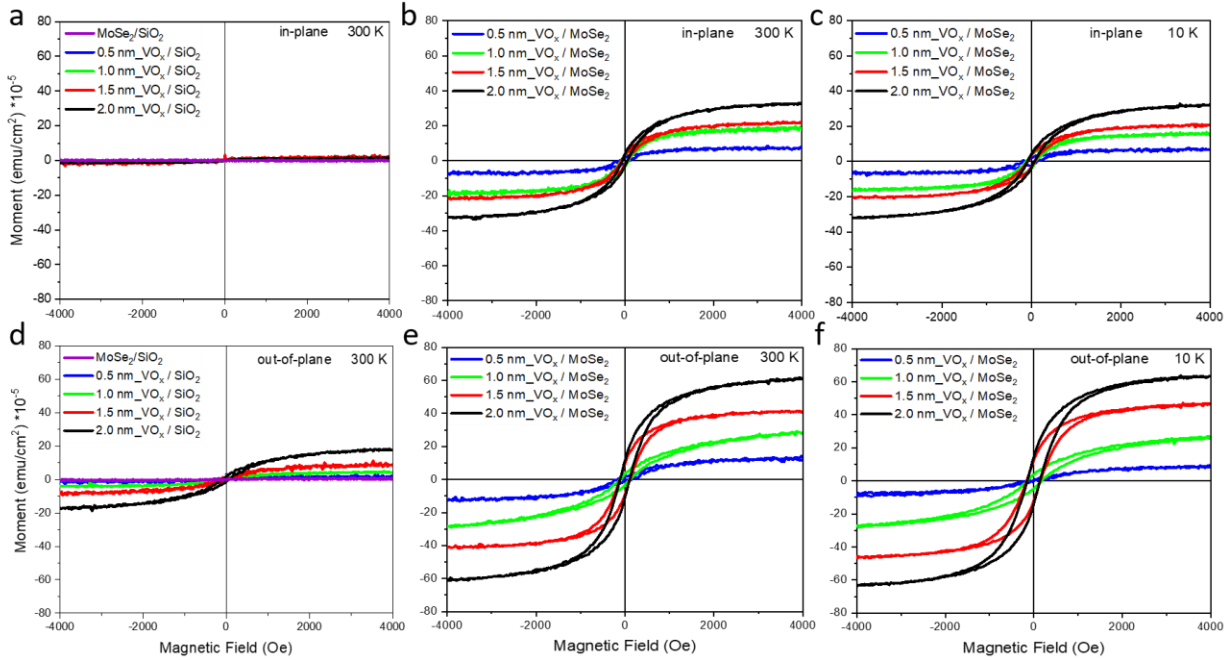


Figure 5. Magnetic properties of the individual constituents and $\text{VO}_x/\text{MoSe}_2$ heterostructures, with the external magnetic field applied in-plane (top panels) and out-of-plane (bottom panels). (a,d) Magnetic response of pristine $\text{MoSe}_2/\text{SiO}_2$ and VO_x/SiO_2 films with different nominal VO_x thicknesses. (b, e) Hysteresis loops measured at 300 K and (c,f) at 10 K for $\text{VO}_x/\text{MoSe}_2/\text{SiO}_2$ heterostructures with different nominal VO_x thicknesses.

Both pristine MoSe_2 and bulk V_2O_5 exhibit negligible calculated magnetic moments, consistent with the nonmagnetic character expected for these reference systems.⁴⁰ In contrast, finite VO_x clusters develop localized magnetic moments upon structural relaxation. As illustrated for V_4O_{10} in Figure 6b,c, relaxation modifies the local coordination of the finite oxide cluster and produces incomplete compensation between positive and negative spin contributions, resulting in a net magnetic moment of approximately $1.20\mu_B$. The corresponding spin-density distribution (Figure 6d) shows that this magnetic polarization is localized within the finite oxide cluster. A substantially larger moment of approximately $7.0\mu_B$ is obtained for the isolated $\text{V}_{10}\text{O}_{22}$ cluster,

further demonstrating that finite vanadium oxide nanostructures can possess intrinsic magnetic moments even though bulk V_2O_5 is nonmagnetic.

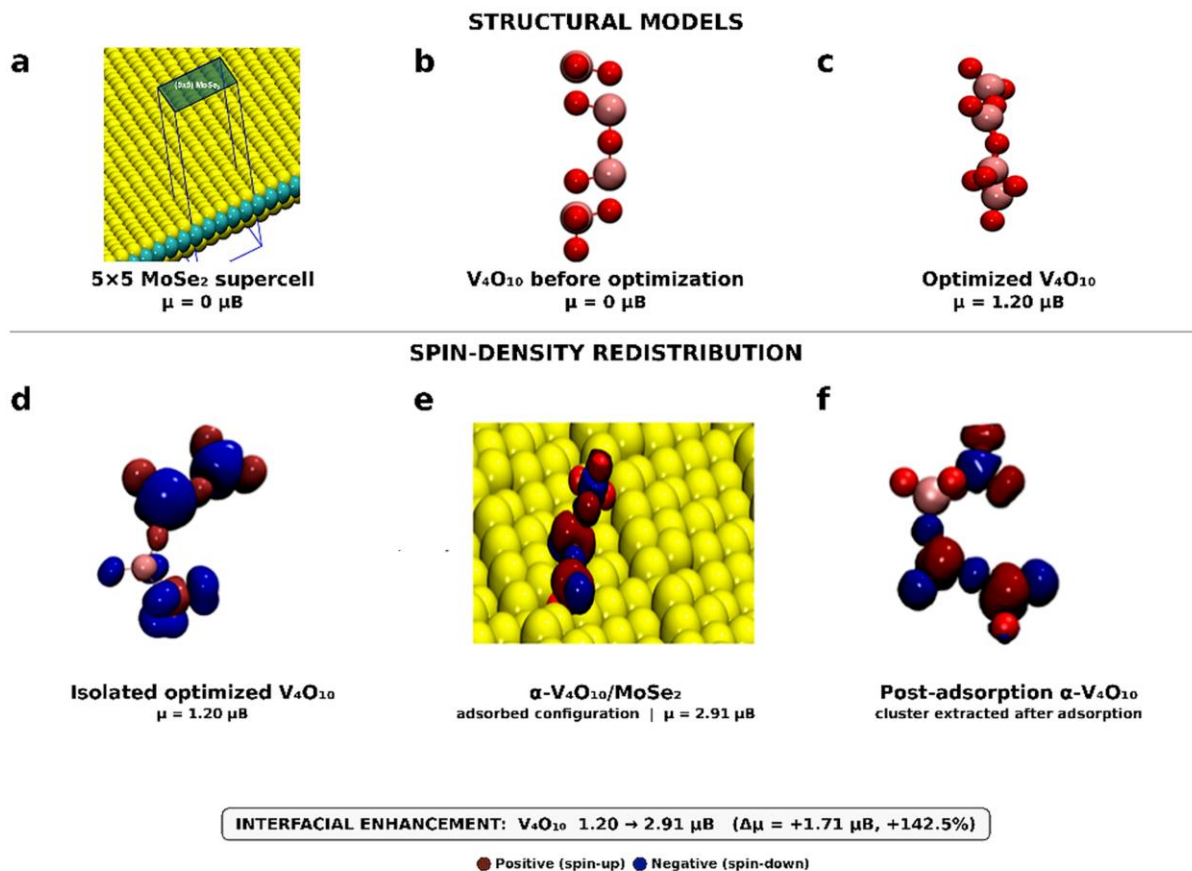


Figure 6. Structural evolution and spin-density redistribution in the $V_4O_{10}/MoSe_2$ system. (a) 5×5 $MoSe_2$ supercell employed to construct the hybrid structures. (b) Initial bulk-derived V_4O_{10} cluster prior to structural relaxation and (c) optimized isolated V_4O_{10} cluster. Structural relaxation results in incomplete compensation between local spin contributions and produces a finite magnetic moment of $1.20\mu_B$. (d) Spin-density distribution of the optimized isolated V_4O_{10} cluster. (e) Spin-density distribution of the α - $V_4O_{10}/MoSe_2$ configuration after adsorption, for which the total magnetic moment increases to $2.91\mu_B$. (f) Spin-density distribution of the V_4O_{10} cluster extracted from the optimized heterostructure while retaining its post-adsorption geometry. Brown and blue isosurfaces represent positive (spin-up) and negative (spin-down) spin densities, respectively.

Interaction with MoSe₂ further modifies this spin-polarized state. As shown in Figure 6e, adsorption of the α -V₄O₁₀ cluster produces a pronounced redistribution of spin density within the oxide, concomitant with an increase in the total magnetic moment from 1.20 to 2.91 μ_B , corresponding to an enhancement of approximately 142.5%. Charge-density analysis further shows substantial electronic redistribution in the interfacial region, involving Se–O bonding, consistent with charge transfer between MoSe₂ and the oxide. Importantly, the enhanced spin polarization remains predominantly localized within V₄O₁₀ rather than being transferred to the MoSe₂ layer itself.

To further distinguish the effect of the relaxed adsorption geometry from the direct electronic contribution of the substrate, the V₄O₁₀ cluster was subsequently extracted from the optimized heterostructure while retaining its post-adsorption geometry (Figure 6f). Comparison of Figures 6d–f therefore allows the spin redistribution associated with structural relaxation and that arising directly from interfacial electronic coupling to be assessed separately. A similar interfacial enhancement is obtained for the larger V₁₀O₂₂ cluster, whose calculated magnetic moment increases from approximately 7.0 to 8.0 μ_B after adsorption. Although the relative enhancement is smaller than for V₄O₁₀, this result indicates that the interfacial modulation of the oxide magnetic state is not restricted to a single cluster size or geometry.

Taken together, these calculations indicate that MoSe₂ does not become substantially magnetic upon contact with VO_x; instead, interfacial electronic reconstruction enhances magnetic moments that already emerge in finite oxide clusters, providing a microscopic explanation for the increased magnetic response observed experimentally in the VO_x/MoSe₂ heterostructures.

The calculated direction of charge transfer also provides a consistent interpretation of the optical measurements. Electron transfer from MoSe₂ to VO_x corresponds to hole doping of the semiconductor and is therefore consistent with an increased contribution from positively charged trions. The PL measurements alone do not distinguish positive from negative trions, so this assignment is made only in conjunction with the independently calculated direction of charge transfer.

The thickness-dependent temperature evolution of magnetic properties was also investigated by VSM. Figure 7 summarizes the out-of-plane saturation magnetization (M_s) and coercive field (H_c). Panels 7a and 7b show M_s and H_c , respectively, as functions of nominal VO_x thickness

between 10 and 350 K, whereas panels 7c and 7d show their temperature dependences for representative 1.5 and 0.5 nm samples.

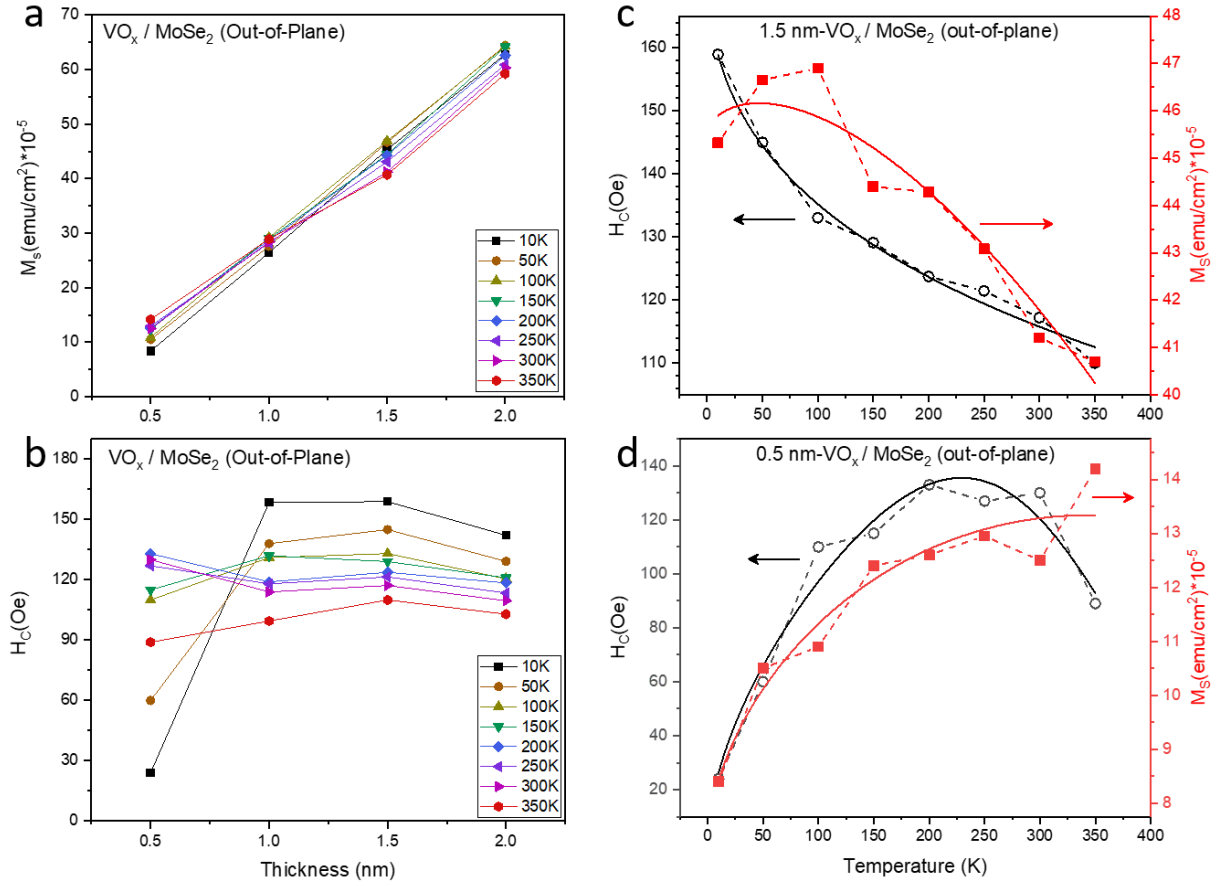


Figure 7. (a,b) Saturation magnetization (M_s) and coercive field (H_c), respectively, as functions of nominal VO_x thickness at different temperatures. (c,d) M_s (squares, right axes) and H_c (circles, left axes) as functions of temperature for nominal VO_x thicknesses of 1.5 and 0.5 nm, respectively. Solid lines are fits using the phenomenological model described in the text.

A pronounced change in the magnetic behavior is observed as the VO_x thickness decreases below approximately 1 nm. The 1.5 nm sample exhibits the conventional temperature dependence expected for a continuous magnetic film. The saturation magnetization decreases gradually from approximately 47×10^{-5} to 41×10^{-5} emu·cm⁻² as the temperature increases from 10 to 350 K, while the coercive field decreases nearly monotonically from approximately 160 to 110 Oe. Such behavior is consistent with the progressive thermal excitation of spin fluctuations together with thermal activation over magnetic anisotropy barriers.

In striking contrast, the ultrathin 0.5 nm sample exhibits the opposite trend. The saturation magnetization increases continuously with increasing temperature, rising from 8.4×10^{-5} to $14.2 \times 10^{-5} \text{ emu} \cdot \text{cm}^{-2}$, while the coercive field increases rapidly between 10 K and approximately 200 K before remaining nearly constant and eventually decreasing at the highest measured temperature. The coexistence of increasing magnetization and increasing coercivity with temperature is highly unusual for conventional ferromagnets and therefore suggests that different collective magnetic dynamics dominate in the ultrathin limit.

These results indicate that the temperature-dependent magnetic response changes qualitatively with oxide morphology. The AFM roughness measurements (Figure 2f) are consistent with discontinuous islands at subnanometer VO_x coverage and a more laterally connected morphology at larger thickness. Accordingly, the contrasting $M_s(T)$ and $H_c(T)$ responses of the 0.5 and 1.5 nm samples cannot be described by ordinary thermal demagnetization alone; they also reflect morphology-dependent reversal dynamics.

We therefore use a phenomenological model to describe the temperature-dependent magnetic properties in Figure 7c,d. The model treats the charge-transfer-enhanced moments identified by DFT as pre-existing magnetic entities once the heterostructure is formed and assigns the contrasting temperature trends to differences in their collective dynamics. In this description, interfacial electronic reconstruction sets the magnetic-moment scale, whereas morphology controls the distribution of anisotropy- and interaction-modified reversal barriers.

This interpretation follows the established framework for fine magnetic particles and nanoparticle assemblies. In the Néel-Brown picture, a single-domain magnetic moment reverses by thermally activated crossing of an anisotropy barrier, with a characteristic relaxation time $\tau = \tau_0 e^{(E/k_B T)}$.^{55, 56} Consequently, whether a moment appears blocked or dynamically responsive depends on the relation between its relaxation time and the experimental observation time. In a real cluster ensemble, variations in cluster size, shape, local anisotropy, defect structure, and orientation produce a distribution of barriers rather than one unique barrier.⁵⁷⁻⁵⁹ Interparticle dipolar or exchange interactions further reshape this energy landscape and can increase, decrease, or broaden the effective barriers governing relaxation and hysteresis.^{58, 60-63} Experimental studies of nanoparticle assemblies have directly shown that interactions and size dispersion modify blocking behavior, relaxation, and coercivity.^{60, 61, 64, 65}

For the discontinuous subnanometer VO_x film, the AFM morphology naturally motivates a distributed-barrier description. At low temperature, a fraction of the cluster moments is trapped in metastable configurations imposed by local anisotropy, disorder, and interactions with neighboring clusters. These moments contribute incompletely to the field-aligned magnetization within the finite field range and timescale of the hysteresis measurement. As temperature increases, a progressively larger fraction of the barrier distribution becomes dynamically accessible, allowing additional pre-existing moments to reach field-favored configurations during the measurement. Thus, the quantity that increases with temperature is the field-accessible contribution of the cluster ensemble, not the number of microscopic moments or the physical connectivity of the islands. This interpretation is consistent with the time-window dependence of blocked-particle magnetism and the evolution of hysteresis when thermally activated reversal becomes accessible.^{55-59, 66}

For the more continuous film, the oxide forms a more strongly connected magnetic network. Most moments can contribute to the macroscopic response already at low temperature, so the dynamically restricted contribution is small. The temperature dependence is therefore dominated by thermal disorder and thermally assisted reversal, producing the conventional reductions in M_s and H_c . The two samples can thus be treated as different limits of the same competition between intrinsic thermal demagnetization and dynamic access to a distributed set of reversal barriers.

In the phenomenological magnetization model, we represent the measured saturation magnetization as, $M_s(T) = M_0(1 - cT^{3/2})[1 + \alpha g(T)]$, where the first factor describes the decrease of the intrinsic magnetic contribution with increasing temperature. The $T^{3/2}$ term is used only as a compact phenomenological representation of low-to-intermediate-temperature spin-wave-like demagnetization. The fitted coefficient c should therefore be interpreted as an effective thermal-demagnetization parameter rather than as a direct measurement of bulk spin-wave stiffness. M_0 is the low-temperature magnetic scale for a given sample and incorporates the amount of oxide, the charge-transfer enhancement identified by DFT, and the population of magnetic defects or clusters.

The second factor in the M_s equation accounts for the additional field-accessible contribution of moments whose response is dynamically restricted at low temperature. Here, $g(T)$ is the dynamically accessible fraction of these moments, and α is the relative magnitude of the

magnetic-moment population whose field-aligned contribution is restricted at low temperature. A small α corresponds to a response that is already largely accessible at low temperature; whereas a large α signifies a substantial cluster contribution that becomes accessible across the measured temperature range.

Within a distributed-barrier picture, the dynamically accessible fraction $g(T)$ can be written conceptually as:

$$g(T) = \int_0^{E_c(T)} P(E) dE, \quad \text{with } E_c(T) = k_B T \ln(\tau_m/\tau_0)$$

where $P(E)$ is the distribution of effective reversal barriers, τ_m is the characteristic experimental observation time, and τ_0 is the attempt time. The threshold $E_c(T)$ follows from the Néel-Brown condition $\tau(E, T) \leq \tau_m$.^{55, 56} Because the present static magnetometry data do not independently determine $P(E)$, τ_m , and τ_0 , we represent the cumulative response over the measured interval by the minimal normalized form: $g(t) = (T/T_{max})^n$. Here, the exponent n is an effective shape parameter for the cumulative barrier distribution sampled by the experiment, and controls how rapidly dynamic accessibility develops with temperature; T_{max} is fixed at the upper experimental limit 350 K. Fixing T_{max} removes an unconstrained activation-temperature parameter and does not introduce an additional fitted energy scale.

The empirical power law used for $g(T)$ is not proposed as a universal microscopic barrier distribution. Rather, it is a minimal-parameter approximation to the monotonic cumulative accessibility expected for a broad barrier ensemble over a finite temperature window. Distributed-barrier and interaction-modified relaxation approaches are widely used in analyses of nanoparticle assemblies.⁵⁷⁻⁶⁵

On the other hand, coercivity H_c is intrinsically sensitive to thermally activated reversal because a hysteresis loop records the field at which metastable magnetic states lose stability within a finite observation time. Thermal activation lowers the switching field relative to its zero-temperature value, as described by Néel-Brown and Sharrock-type treatments and by subsequent finite-temperature micromagnetic analyses.^{55, 56, 66-68} For the continuous film, the observed monotonic softening is represented by $H_c(T) = H_{c0} e^{-(T/T_s)^r}$.

Here H_{c0} is the extrapolated low-temperature coercive scale, T_s is an effective thermal-softening scale—not a transition temperature—and r controls the curvature of the softening. The stretched exponential is used phenomenologically to represent a distribution of reversal processes rather than a single barrier.

For the clustered film, the same distribution of anisotropy barriers that limits magnetic alignment also determines which clusters can switch their magnetization during the hysteresis cycle. In other words, the barrier distribution determines both the field-aligned magnetic response (how many cluster moments can access field-favored configurations) and the dynamic magnetic response (how rapidly and at what field the clusters can switch). The coercive field for the clustered film is therefore written as,

$$H_c(T) = [H_{base} + H_1 g(T)] e^{-(T/T_s)^r}$$

The bracketed term describes the progressive involvement of dynamically accessible cluster moments in hysteretic reversal. H_{base} is the residual low-temperature coercive contribution and H_1 measures the additional coercive scale associated with the cluster population entering the experimental time window. The exponential factor describes the competing thermal reduction of effective reversal barriers. At low and intermediate temperatures, the growth of $g(T)$ can dominate and increase H_c ; at higher temperature, thermal softening dominates, producing a maximum followed by a decrease. Nonmonotonic coercivity can arise in size-distributed or interacting nanoparticle ensembles when different thermally activated reversal contributions compete, and numerical treatments of distributed assemblies reproduce such behavior.^{64, 69}

In Figure 7c,d, the M_s data for both samples were fitted simultaneously using a common thermal-demagnetization coefficient c and a common exponent n , while M_0 and α were allowed to vary with morphology. The joint fit gives $c = (2.69 \pm 0.66) \times 10^{-5} \text{ K}^{-3/2}$ and $n = 0.569 \pm 0.269$. For the 1.5 nm sample, $M_0 = (45.49 \pm 0.88) \times 10^{-5} \text{ emu cm}^{-2}$ and $\alpha = 0.074 \pm 0.059$, with $R^2 = 0.923$. The small α indicates that nearly the entire measurable moment is accessible at low temperature; the fitted response is consequently governed mainly by conventional thermal demagnetization and reproduces the gradual decrease in M_s above approximately 100 K.

For the 0.5 nm clustered film, $M_0 = (7.26 \pm 1.76) \times 10^{-5} \text{ emu cm}^{-2}$ and $\alpha = 1.23 \pm 0.47$, with $R^2 = 0.921$. The much larger α is the central result of the fit: the dynamically restricted

contribution is comparable to the low-temperature magnetic scale, consistent with the broad distribution of barriers expected for the island morphology. The increase in $g(T)$ overcompensates the intrinsic thermal-demagnetization term through most of the measured interval, yielding the observed increase in M_s . The common n value provides a compact phenomenological description in which the two morphologies differ primarily in the magnitude of the dynamically restricted population rather than requiring unrelated temperature functions.

The continuous-film coercivity is reproduced by $H_{c0} = 174.7 \pm 8.3$ Oe, $T_s = 2355 \pm 535$ K, and $r = 0.431 \pm 0.092$, with $R^2 = 0.990$. The large effective T_s and small r produce broad, gradual softening, consistent with a connected magnetic film containing a distribution of reversal barriers. Importantly, T_s is an empirical scale parameter of a stretched-exponential description rather than a physical transition temperature. Its large value reflects the gradual coercive softening observed over the experimental temperature interval and should not be interpreted as a transition or blocking temperature. The same data can be described equivalently over the measured range by a conventional sublinear power-law dependence of the form $H_c = H_0 - A \cdot T^p$.

The clustered-film fit gives $H_{base} \approx 0$, $H_l = 199.3 \pm 31.1$ Oe, $T_s = 374.7 \pm 16.6$ K, and $r = 3.94 \pm 1.49$, with $R^2 = 0.963$. Here, the $H_l g(T)$ term initially raises H_c as more cluster moments become involved in thermally assisted switching and contribute to the hysteresis loop. At higher temperature, the rapidly increasing thermal-softening term becomes dominant and produces the downturn observed at 350 K. The model therefore reproduces both the rise and subsequent decrease using two physically motivated effects acting on the same barrier ensemble.

Taken together, the fitted parameters support a consistent morphology-dependent picture. Charge transfer from MoSe₂ enhances the local magnetic moments, while oxide morphology determines whether those moments form an already accessible connected response or a clustered ensemble with a broad distribution of effective barriers. The more continuous film is dominated by conventional thermal demagnetization and coercive softening. In the island film, thermally assisted access to metastable cluster configurations initially increases both the field-aligned magnetization and the population participating in magnetization switching; at the highest temperatures, thermal softening reduces H_c . This description links the AFM morphology, DFT charge-transfer result, and temperature-dependent hysteresis without assigning the anomalous temperature dependence to charge transfer.

Conclusions

We demonstrate that ultrathin $\text{VO}_x/\text{MoSe}_2$ heterostructures exhibit enhanced magnetism governed by two complementary effects: interfacial electronic reconstruction and oxide morphology. Structural and spectroscopic measurements show a progression from discontinuous VO_x islands toward more continuous films while preserving the underlying MoSe_2 lattice. XPS indicates negligible substitutional V incorporation into MoSe_2 and reveals a mixed-valence, nonstoichiometric VO_x overlayer. First-principles calculations show that finite VO_x clusters already possess magnetic moments and that interaction with MoSe_2 further enhances those moments through interfacial charge redistribution.

The thinnest films display an unusual increase in magnetization and a nonmonotonic coercivity with temperature, in contrast to the conventional thermal demagnetization and coercive softening of the more continuous film. A distributed-barrier phenomenological model captures this crossover by treating morphology as the factor that controls the collective accessibility and reversal dynamics of the pre-existing, charge-transfer-enhanced moments. The model therefore separates the roles of interface and morphology: electronic reconstruction enhances the local magnetic moment, whereas morphology governs its temperature-dependent collective response.

These results establish $\text{VO}_x/\text{MoSe}_2$ as an oxide/TMD platform in which magnetic functionality can be tuned without relying on substitutional doping of the semiconductor. More broadly, they show how interfacial charge transfer and nanoscale morphology can provide complementary control parameters for magnetic behavior in atomically thin heterostructures, with potential relevance to spintronic, magneto-optical, and multifunctional devices.

Experimental Section

CVD Synthesis of MoSe_2 : Si substrates with a 285 nm SiO_2 layer were cleaned sequentially with acetone, isopropanol, and deionized water before growth. Growth was performed in a 1 in. diameter quartz-tube reactor. As shown in Figure 1a, the SiO_2/Si substrates were positioned downstream in a 700-800 °C zone, while MoSe_2 powder (99.9%, Sigma-Aldrich) in a ceramic boat was placed upstream at the furnace center. The substrates were located 6-10 cm from the solid source. The furnace was heated to approximately 1040 °C over 60 min under N_2 flow (200 sccm), with the source and substrates initially outside the hot zone. When the furnace reached

approximately 1035 °C, the source and substrates were moved to their growth positions by sliding the quartz tube. Simultaneously, water vapor was introduced by diverting the N₂ carrier gas through a bubbler containing 2 mL of deionized water at 24 °C. The N₂/H₂O carrier flow was maintained for 8 min. After growth, the gas was switched to Ar containing 5% H₂ (200 sccm) for 4 min as a termination step.

High-Vacuum Thermal Evaporation of VO_x: Ultrathin VO_x films were thermally evaporated onto CVD-grown MoSe₂ using an Angstrom Engineering high-vacuum evaporator equipped with a cryogenic pumping system. V₂O₅ was used as the evaporation source. The films were deposited at approximately 0.04 Å s⁻¹, and the nominal thickness was monitored with an INFICON quartz crystal sensor. Before deposition, the chamber was evacuated to approximately 10⁻⁷ Torr; during deposition, the pressure increased to approximately 10⁻⁶ Torr. The samples were maintained at room temperature.

Raman and PL Spectroscopies: Raman and photoluminescence measurements were performed with a confocal Raman microscope (LabRAM HR Evolution, Horiba Scientific) in backscattering geometry using 532 nm laser excitation. Raman and PL maps were acquired with a 50× objective while the sample remained stationary and the laser was scanned using the DuoScan™ imaging mode.

XPS Measurements: XPS spectra of as-prepared MoSe₂ and MoSe₂ coated with an ultrathin VO_x layer were acquired in an ultrahigh-vacuum chamber equipped with a nonmonochromated dual-anode X-ray source and a Scienta R3000 hemispherical energy analyzer. The spectra reported here were collected using Al Kα radiation. Binding energies were calibrated to the SiO₂ Si 2p peak at 103.6 eV.

Magnetic Measurements – Vibrating Sample Magnetometry: Field- and temperature-dependent magnetic measurements were performed using the VSM option of a Quantum Design Physical Property Measurement System (PPMS). M-H loops were measured with the magnetic field applied both parallel and perpendicular to the sample plane, as indicated for the corresponding data sets, over the 10-350 K range discussed in this work. Temperature-dependent M–H loops were acquired sequentially upon cooling from 350 to 10 K. After each hysteresis measurement, the magnetic field was returned to zero and the sample was cooled in zero applied field to the next measurement temperature before acquiring the subsequent M–H loop. Saturation magnetization (M_s) was determined from the high-field plateau of the background-subtracted M–H loops, while the

coercive field (H_c) was obtained from the zero-magnetization crossing of the hysteresis loop. Magnetization is reported per unit area. To estimate the actual MoSe_2 -covered area for the discontinuous CVD films, large-area optical images were acquired using the collage-scanning mode of the LabRAM HR Evolution and analyzed with ImageJ; the resulting surface-coverage fraction was used in the areal normalization. Control measurements performed on clean SiO_2 substrates and pristine $\text{MoSe}_2/\text{SiO}_2$ samples using the same mounting and measurement procedures showed no detectable hysteresis within the experimental sensitivity. Together with the systematic VO_x -thickness and temperature dependence observed only in the heterostructures, these controls argue against a substrate-, holder-, or handling-related ferromagnetic contribution.

Density Functional Theory Section: We performed first-principles calculations using density functional theory with a Hubbard correction (DFT+U).⁷⁰ Calculations were carried out with the SIESTA code^{71, 72} using the Perdew-Burke-Ernzerhof (PBE) generalized-gradient approximation (GGA) functional⁷³ and a double- ζ polarized (DZP) basis set. A mesh cutoff of 250 Ry, a Γ -centered $1 \times 1 \times 1$ Monkhorst-Pack k-point grid⁷⁴ and a Hubbard parameter $U = 4.0$ eV for V were used.⁷⁵ A hexagonal $5 \times 5 \times 1$ MoSe_2 supercell containing 75 atoms was first constructed using a previously optimized structure, with $a = b = 16.75$ Å, $c = 30.00$ Å, $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$. The optimized parameters were consistent with literature values,⁷⁶⁻⁷⁹ supporting the computational setup. Two finite vanadium oxide clusters, V_4O_{10} and $\text{V}_{10}\text{O}_{22}$, were then constructed from the bulk V_2O_5 structure and placed on the MoSe_2 supercell to examine their interfacial interaction. The V_4O_{10} cluster contains the 14 atoms of one bulk V_2O_5 unit cell but is treated as a finite, nonperiodic cluster, with approximate dimensions of $2.54 \times 9.22 \times 5.34$ Å³. The $\text{V}_{10}\text{O}_{22}$ cluster is a 32-atom slab-like structure derived from bulk V_2O_5 cleaved along the (001) direction, with approximate dimensions of $8.67 \times 3.30 \times 10.18$ Å³. Hybrid structures were constructed by placing each cluster on the MoSe_2 supercell. These finite-cluster models are intended to represent local $\text{VO}_x/\text{MoSe}_2$ interactions relevant to the early stages of ultrathin-film growth rather than the full morphology of the experimental films.

Acknowledgements

H.R.G. acknowledges support from the National Science Foundation Grant No. NSF-ECCS-2430871, and from Nanobase Inc. Award: “Optical Properties of New Two-Dimensional Materials”. H.R.G. thanks Prof. Matthias Batzill for kindly granting access to his laboratory

facilities. B.I. and R.B.O. thank CNPq (process numbers 194155/2025-0 and 151043/2024-8, respectively). B.I. and G.S.L.F. thank FAPESP (process numbers 2024/11016-0 and 2024/03413-9, respectively), D.S.G. acknowledges the Center for Computing in Engineering and Sciences at UNICAMP for financial support through the FAPESP/CEPID grant 2013/08293-7. The authors also thank the Coaraci Supercomputer for computer time (FAPESP grant 2019/17874-0) and the Center for Computing in Engineering and Sciences at UNICAMP (FAPESP grant 2013/08293-4). D. S. G. also acknowledges support from INEO/CNPq and FAPESP grant 2025/27044-5.

Conflict of interest

The authors declare no conflict of interest.

Author Contributions

Y.W. and N.K. conducted synthesis experiments as well as Raman and PL spectroscopies with F.A.N. assistance. K.L. conducted XPS characterization. Y. W. performed VSM and AFM measurements, processed and analyzed the data with H.R.G. input. J.P. provided complementary AFM data. X.J. contributed with discussions and resources (high-vacuum thermal evaporation). R.B.O., B.I., G.S.L.F. and D.S.G. performed DFT calculations. H.R.G., Y.W. and N.K. conceived the project and wrote the manuscript with contributions from the other authors. All the authors read, commented on the manuscript, and gave their approval for the final manuscript.

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